

Chapter 4

4.1 $\Sigma \vec{F} = 0$ is a three equation set:

$$\Sigma F_x = 0; \Sigma F_y = 0; \Sigma F_z = 0.$$

$$\Sigma F_x = 0$$

Face BEFC:

$$(\sigma_x + \frac{\partial \sigma_x}{\partial x} dx) dy dz$$

$$AHGD: (-\sigma_x) dy dz$$

$$CFGD: (\tau_{xy} + \frac{\partial \tau_{xy}}{\partial y} dy) dx dz$$

$$ABEH: (-\tau_{xy}) dx dz$$

$$ABCD: (\tau_{zx} + \frac{\partial \tau_{zx}}{\partial z} dz) dx dy$$

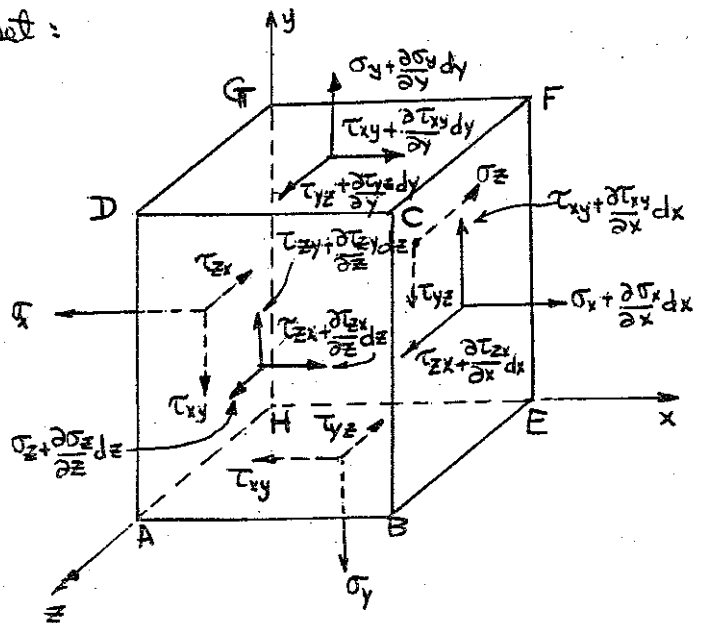
$$EFGH: (-\tau_{zx}) dx dy$$

$$\therefore (\sigma_x + \frac{\partial \sigma_x}{\partial x} dx) dy dz - \sigma_x dy dz + (\tau_{xy} + \frac{\partial \tau_{xy}}{\partial y} dy) dx dz - \tau_{xy} dx dz + (\tau_{zx} + \frac{\partial \tau_{zx}}{\partial z} dz) dx dy - \tau_{zx} dx dy = 0$$

$$\text{i.e. } (\frac{\partial \sigma_x}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} + \frac{\partial \tau_{zx}}{\partial z})(dx dy dz) = 0 \quad \text{i.e. } (\frac{\partial \sigma_x}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} + \frac{\partial \tau_{zx}}{\partial z}) = 0$$

$$\text{Similarly: } \Sigma F_y = 0 : \frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \sigma_y}{\partial y} + \frac{\partial \tau_{yz}}{\partial z} = 0$$

$$\Sigma F_z = 0 : \frac{\partial \tau_{zx}}{\partial x} + \frac{\partial \tau_{yz}}{\partial y} + \frac{\partial \sigma_z}{\partial z} = 0.$$



4.2 Refer to figure for prob 4.1.

In the equation $\Sigma F_x = 0$ we have another term on the left hand side = $X(dx dy dz)$

$$\text{Thus } \Sigma F_x = 0 : \frac{\partial \sigma_x}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} + \frac{\partial \tau_{zx}}{\partial z} + X = 0$$

$$\Sigma F_y = 0 : \frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \sigma_y}{\partial y} + \frac{\partial \tau_{yz}}{\partial z} + Y = 0$$

$$\Sigma F_z = 0 : \frac{\partial \tau_{zx}}{\partial x} + \frac{\partial \tau_{yz}}{\partial y} + \frac{\partial \sigma_z}{\partial z} + Z = 0.$$

4.3 $\Sigma \vec{F} = 0 \rightarrow 1) \Sigma F_x = 0, 2) \Sigma F_y = 0$

$\Sigma F_x = 0 :$

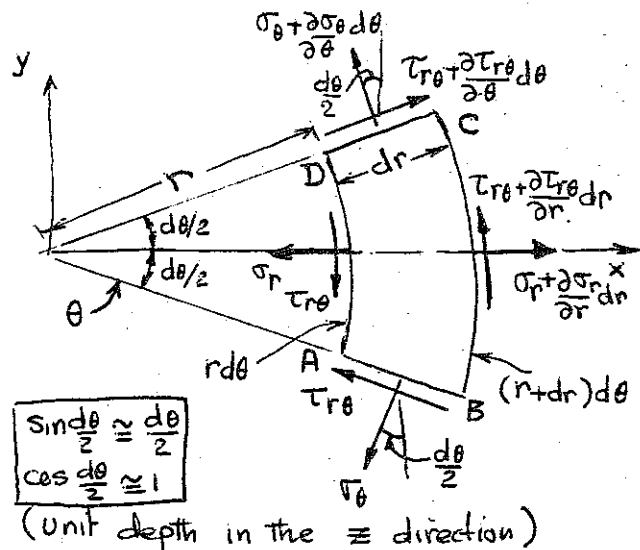
face BC :

$(\sigma_r + \frac{\partial \sigma_r}{\partial r} dr)(r+dr)d\theta$

AD : $(-\sigma_r)(rd\theta)$

AB : $(-\tau_{r\theta})(dr) - (\sigma_\theta)(dr)(\frac{d\theta}{2})$

CD : $(\tau_{r\theta} + \frac{\partial \tau_{r\theta}}{\partial \theta} d\theta) dr$
 $- (\sigma_\theta + \frac{\partial \sigma_\theta}{\partial \theta} d\theta)(dr)(\frac{d\theta}{2})$



Thus : $(\sigma_r + \frac{\partial \sigma_r}{\partial r} dr)(r+dr)d\theta - \sigma_r rd\theta - \tau_{r\theta} dr - \sigma_\theta \frac{dr d\theta}{2} + (\tau_{r\theta} + \frac{\partial \tau_{r\theta}}{\partial \theta} d\theta) dr$
 $- (\sigma_\theta + \frac{\partial \sigma_\theta}{\partial \theta} d\theta) \frac{dr d\theta}{2} = 0.$

ie $\sigma_r rd\theta + \sigma_r dr d\theta + r \frac{\partial \sigma_r}{\partial r} dr d\theta + \frac{\partial \sigma_r}{\partial r} (dr)^2 d\theta - \sigma_r rd\theta - \tau_{r\theta} dr - \sigma_\theta \frac{dr d\theta}{2}$
 $+ \tau_{r\theta} dr + \frac{\partial \tau_{r\theta}}{\partial \theta} d\theta dr - \sigma_\theta \frac{dr d\theta}{2} - \frac{\partial \sigma_\theta}{\partial \theta} dr (d\theta)^2 = 0.$

Retaining only the second order quantities, we obtain :

$(\sigma_r + r \frac{\partial \sigma_r}{\partial r} - \frac{\sigma_\theta}{2} + \frac{\partial \tau_{r\theta}}{\partial r} - \frac{\sigma_\theta}{2}) = 0$ ie $\boxed{\frac{\partial \sigma_r}{\partial r} + \frac{1}{r} \frac{\partial \tau_{r\theta}}{\partial \theta} + \frac{\sigma_r - \sigma_\theta}{r} = 0}$

Similarly for $\Sigma F_y = 0 :$

$\boxed{\frac{\partial \tau_{r\theta}}{\partial r} + \frac{1}{r} \frac{\partial \sigma_\theta}{\partial \theta} + \frac{2\tau_{r\theta}}{r} = 0}$

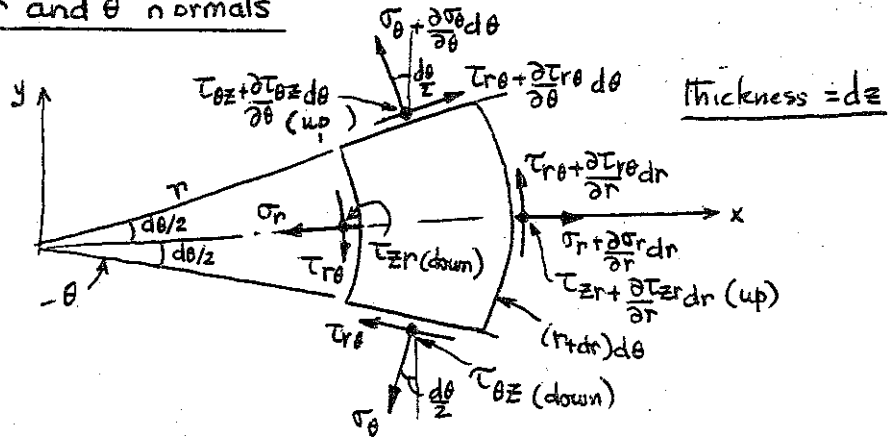
4.4 An elementary volume will have three pairs of faces with normals in the r, θ and z directions respectively.

Considering each pair in turn we will write down the forces in the x, y, z directions

Finally we will write $\Sigma F_x = 0, \Sigma F_y = 0, \Sigma F_z = 0.$

4.4 continued

1) Faces with r and θ normals



$$F_x = \left[\left(\sigma_r + \frac{\partial \sigma_r}{\partial r} dr \right) (r+dr) d\theta - \sigma_r r d\theta - \tau_{r\theta} dr - \sigma_\theta dr \frac{d\theta}{2} + \left(\tau_{r\theta} + \frac{\partial \tau_{r\theta}}{\partial \theta} d\theta \right) dr - \left(\sigma_\theta + \frac{\partial \sigma_\theta}{\partial \theta} d\theta \right) dr \frac{d\theta}{2} \right] dz \approx \left(r \frac{\partial \sigma_r}{\partial r} + \sigma_r - \sigma_\theta + \frac{\partial \tau_{r\theta}}{\partial \theta} \right) (dr d\theta dz) \quad (\text{dropping higher order quantities})$$

$$F_y = \left[\left(\sigma_\theta + \frac{\partial \sigma_\theta}{\partial \theta} d\theta \right) dr + \left(\tau_{r\theta} + \frac{\partial \tau_{r\theta}}{\partial \theta} d\theta \right) dr \frac{d\theta}{2} - \sigma_\theta dr + \tau_{r\theta} dr \frac{d\theta}{2} + \left(\tau_{r\theta} + \frac{\partial \tau_{r\theta}}{\partial r} dr \right) (r+dr) d\theta - \tau_{r\theta} (r d\theta) \right] dz \approx \left(\frac{\partial \sigma_\theta}{\partial \theta} + \tau_{r\theta} + r \frac{\partial \tau_{r\theta}}{\partial r} + \tau_{r\theta} \right) \left(\frac{r d\theta}{2} \right) dz$$

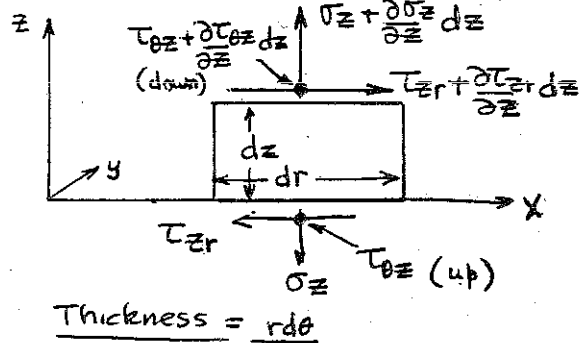
$$F_z = \left[\left(\tau_{r\theta} + \frac{\partial \tau_{r\theta}}{\partial r} dr \right) (r+dr) d\theta - \tau_{r\theta} r d\theta + \left(\tau_{\theta z} + \frac{\partial \tau_{\theta z}}{\partial \theta} d\theta \right) dr - \tau_{\theta z} dr \right] dz \approx \left(r \frac{\partial \tau_{r\theta}}{\partial r} + \tau_{r\theta} + \frac{\partial \tau_{\theta z}}{\partial \theta} \right) (dr d\theta dz)$$

2) Faces with z normals

$$F_x = \left[\left(\tau_{zr} + \frac{\partial \tau_{zr}}{\partial z} dz \right) dr - \tau_{zr} dr \right] r d\theta = \left(r \frac{\partial \tau_{zr}}{\partial z} \right) (dr d\theta dz)$$

$$F_y = \left[\left(\tau_{\theta z} + \frac{\partial \tau_{\theta z}}{\partial z} dz \right) dr - \tau_{\theta z} dr \right] r d\theta = \left(r \frac{\partial \tau_{\theta z}}{\partial z} \right) (dr d\theta dz)$$

$$F_z = \left[\left(\sigma_z + \frac{\partial \sigma_z}{\partial z} dz \right) dr - \sigma_z dr \right] r d\theta = \left(r \frac{\partial \sigma_z}{\partial z} \right) (dr d\theta dz)$$



4.4 cont'd

Thus

$$\Sigma F_x = 0 : \left(r \frac{\partial \sigma_r}{\partial r} + \sigma_r - \sigma_\theta + \frac{\partial \tau_{r\theta}}{\partial \theta} + r \frac{\partial \tau_{\theta r}}{\partial r} \right) (dr d\theta dz) = 0$$

$$\text{i.e. } \frac{\partial \sigma_r}{\partial r} + \frac{1}{r} \frac{\partial \tau_{r\theta}}{\partial \theta} + \frac{\partial \tau_{\theta r}}{\partial r} + \frac{(\sigma_r - \sigma_\theta)}{r} = 0$$

$$\Sigma F_y = 0 : \left(\frac{\partial \sigma_\theta}{\partial \theta} + \tau_{r\theta} + r \frac{\partial \tau_{r\theta}}{\partial r} + \tau_{\theta r} + r \frac{\partial \tau_{\theta z}}{\partial z} \right) (dr d\theta dz) = 0$$

$$\text{i.e. } \frac{\partial \tau_{r\theta}}{\partial r} + \frac{1}{r} \frac{\partial \sigma_\theta}{\partial \theta} + \frac{\partial \tau_{\theta z}}{\partial z} + \frac{2\tau_{r\theta}}{r} = 0$$

$$\Sigma F_z = 0 : \left(r \frac{\partial \tau_{zr}}{\partial r} + \tau_{zr} + \frac{\partial \tau_{\theta z}}{\partial \theta} + r \frac{\partial \sigma_z}{\partial z} \right) (dr d\theta dz) = 0$$

$$\text{i.e. } \frac{\partial \tau_{zr}}{\partial r} + \frac{1}{r} \frac{\partial \tau_{\theta z}}{\partial \theta} + \frac{\partial \sigma_z}{\partial z} + \frac{\tau_{zr}}{r} = 0$$

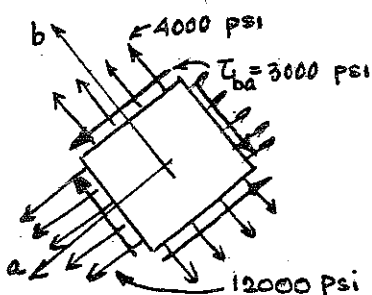
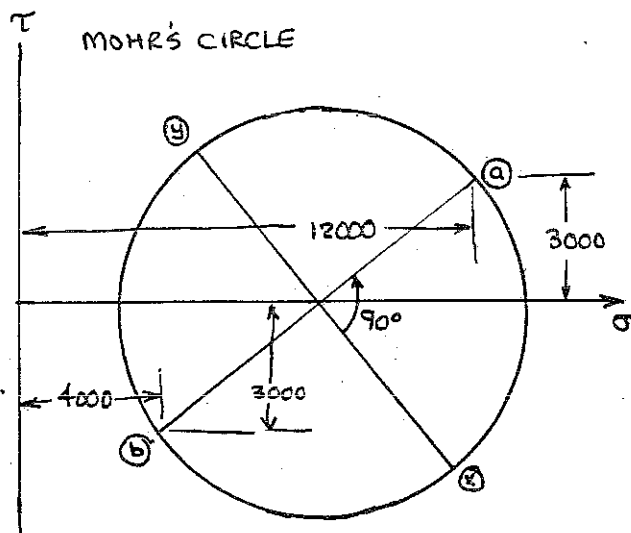
4.5 Reversing the direction of the 'a' axis is equivalent to rotating it 180° in the physical plane. This corresponds to a $2(180) = 360^\circ$ rotation in Mohr's circle, which leaves the point (a) unchanged.

Reversed 'a' axis:

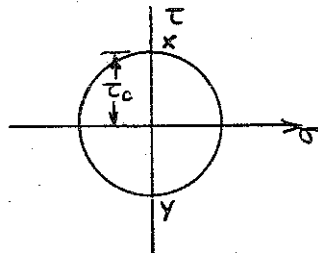
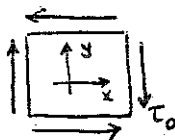
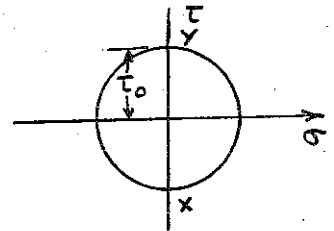
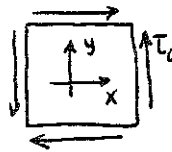
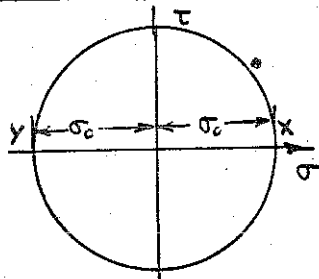
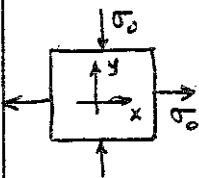
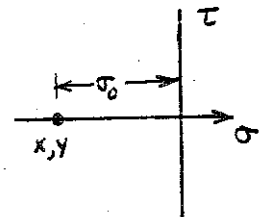
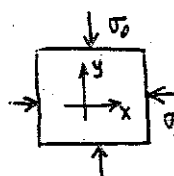
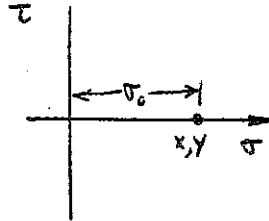
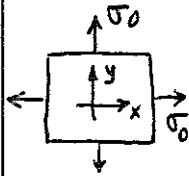
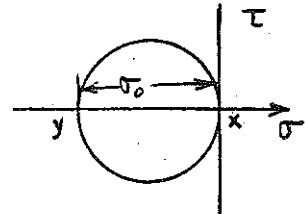
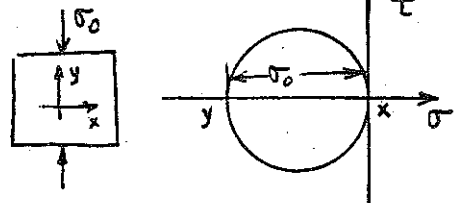
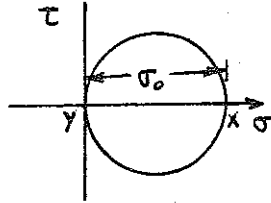
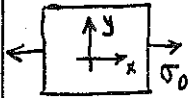
'b' is now clockwise axis

On Mohr's circle, 'b' lies below the σ axis. Therefore, τ_{ba} is positive, and $\tau_{ba} = +3000$ psi.

The resulting stress picture is shown on the right. It can be seen that it is identical to that shown in Fig 4.19c

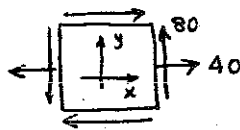


4.6



4.7

a)

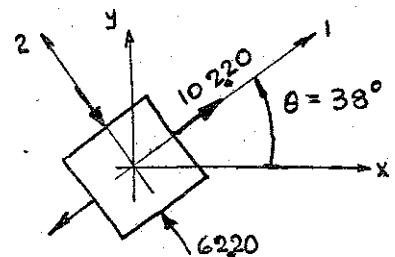
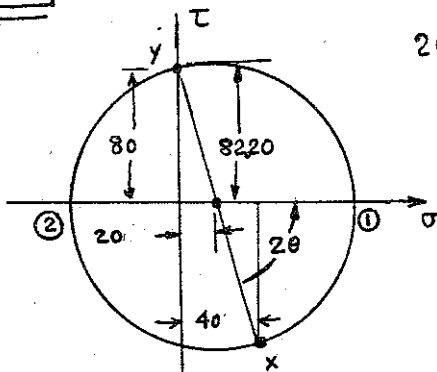


MN/m²

$$\sigma_1 = 10220$$

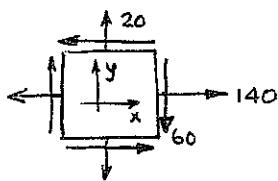
$$\sigma_2 = -6220$$

$$2\theta = 76^\circ \therefore \theta = 38^\circ$$



4.7 cont'd

b)

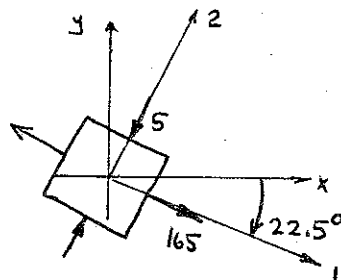
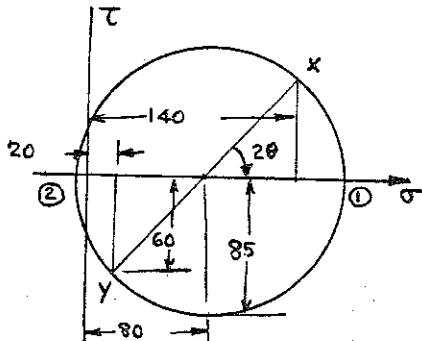


MN/m²

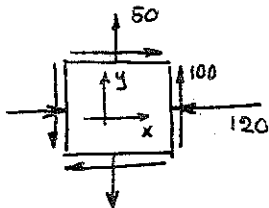
$$\theta = \frac{1}{2} \tan^{-1} \left(\frac{6000}{80} \right) = 22.5^\circ$$

$$\sigma_1 = 80 + 85 = 165$$

$$\sigma_2 = 80 - 85 = -5$$



c)

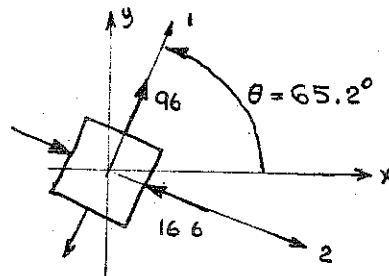
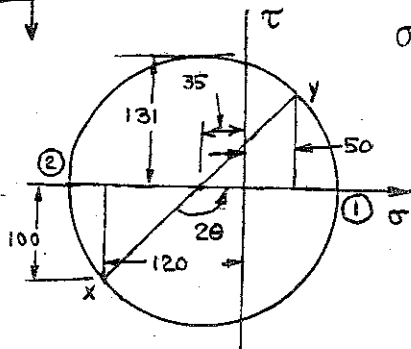


MN/m²

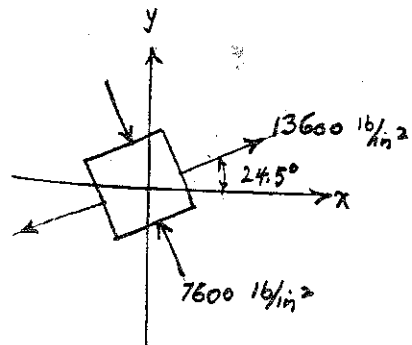
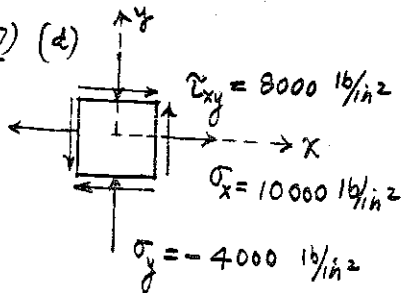
$$\theta = \frac{1}{2} \tan^{-1} \left(\frac{100}{85} \right) = 65.2^\circ$$

$$\sigma_1 = -35 + 131 = +96$$

$$\sigma_2 = -35 - 131 = -166$$



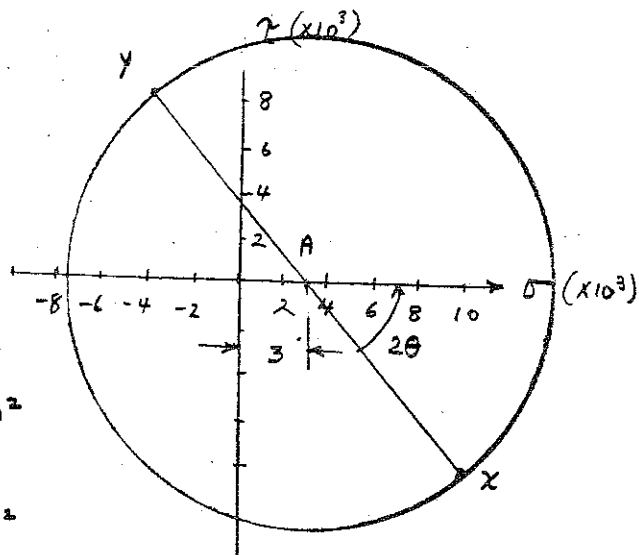
4-7) (d)



$$\sigma_1 = 3000 + AX = 13600 \text{ lb/in}^2$$

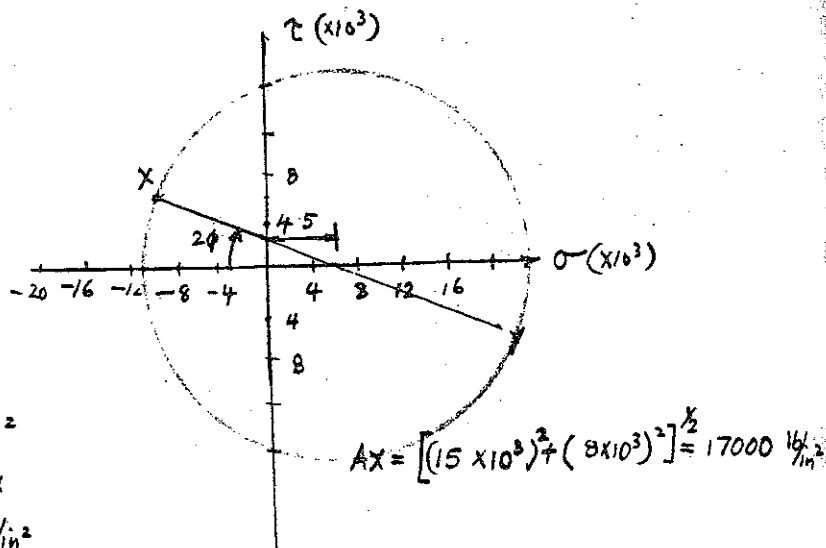
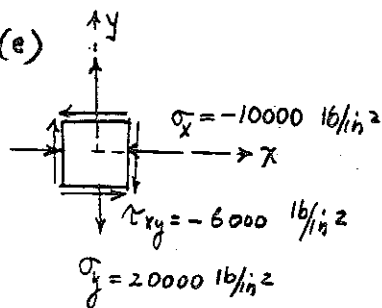
$$\sigma_2 = 3000 - AX = -7600 \text{ lb/in}^2$$

$$2\theta = \tan^{-1} \frac{8000}{7000} \rightarrow \theta = 24.5^\circ$$



$$AX = \left[(8 \times 10^3)^2 + (7 \times 10^3)^2 \right]^{1/2} = 10600 \text{ lb/in}^2$$

(4-7)(e)



$$\begin{aligned}\sigma_2 &= +5000 + Ax \\ &= 22000 \text{ lb/in}^2 \\ \sigma_1 &= +5000 - Ax \\ &= -12000 \text{ lb/in}^2\end{aligned}$$

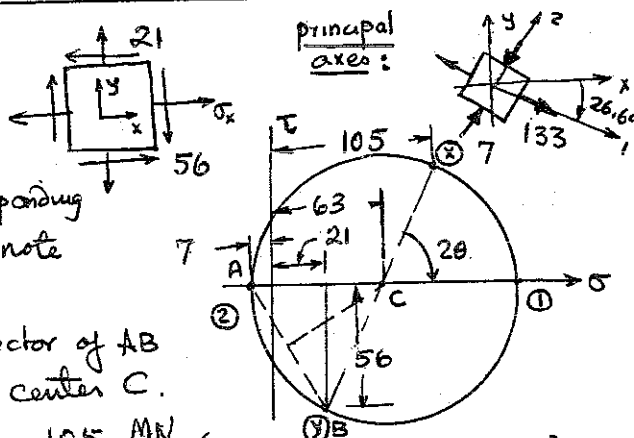
$$\phi = \frac{1}{2} \tan^{-1} \frac{6000}{15000} \rightarrow \phi = 11^\circ$$

4.8

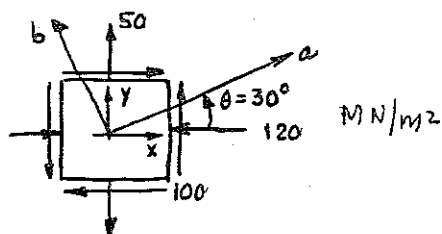
Point A corresponding to the lowest principal stress is located by laying off $\sigma = -7 \frac{\text{MN}}{\text{m}^2}$. Point B, corresponding to 'y' is located as usual (note that τ_{xy} is negative).

Join AB. Perpendicular bisector of AB intersects the σ axis at the center C.

Thus, $\theta = 26.6^\circ$; $\sigma_1 = 133 \frac{\text{MN}}{\text{m}^2}$ $\sigma_x = 105 \frac{\text{MN}}{\text{m}^2}$ (See figure at upper right)



4.9



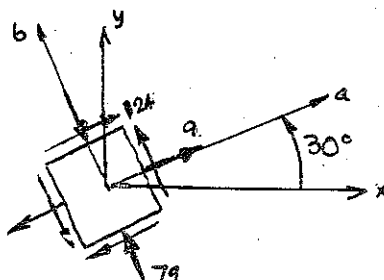
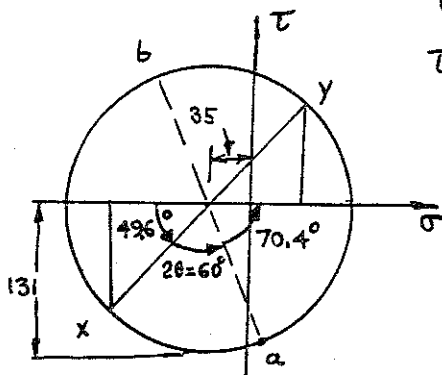
From Mohr's circle:

$$\sigma_a = -35 + 131 \cos(70.4^\circ) \\ = -35 + 44 = 9$$

$$\sigma_b = -35 - 44 = -79$$

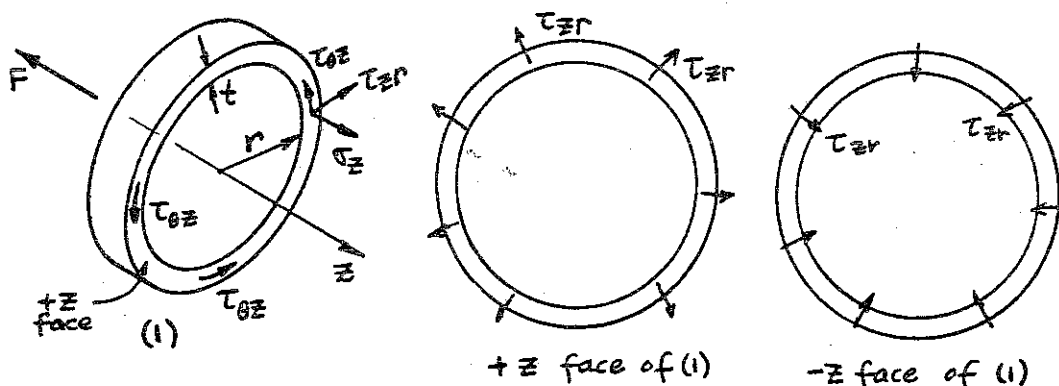
$$\tau_{ab} = \text{positive} = 131 \sin(70.4^\circ) \\ = 124$$

STRESSES,
MN/m²



4.10

Consider, first, the element (1) shown below:



+z face: Circular symmetry: stresses independent of θ . Hence

$\tau_{zr} = \text{constant}$ and suppose it points outwards, as shown.

Now at a sufficient distance away from the point of application of F , the stresses will be independent of z (St. Venant's principle). Therefore, on the -z face, τ_{zr} will have the same magnitude but opposite sign (action and reaction). However, since the choice of the $+z$ direction is arbitrary, we should see the same physical stress state whether we look in at the

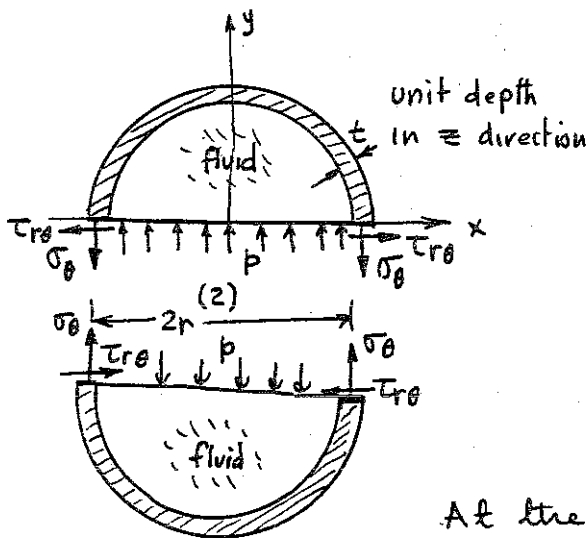
4.10 (cont'd)

element from the $+z$ or the $-z$ directions. This can only be so if $\tau_{zr} = 0$.

EQUILIBRIUM : $\Sigma F_z = 0$: $F = (\sigma_z) 2\pi r t \quad \therefore \sigma_z = \frac{F}{2\pi r t}$

$\Sigma M_z = 0$: $\tau_{\theta z} (2\pi r t) r = 0 \quad \therefore \tau_{\theta z} = 0$

Now consider the element (2) below :



A symmetry argument, similar to that used for element (1) shows that $\tau_{r\theta} = 0$.

EQUILIBRIUM

$\Sigma F_r = 0$: $2[\sigma_\theta(t)(1)] = p(2r)(1)$

$\therefore \sigma_\theta = \frac{pr}{t}$

At the inside surface, $\sigma_r = -p$
at the outside surface, $\sigma_r = 0$

$\therefore$ if t is small it seems reasonable to say that $-p \leq \sigma_r \leq 0$ in the wall. If $(r/t) \gg 1$ then $\sigma_r \ll \sigma_\theta$ and σ_r may be neglected.

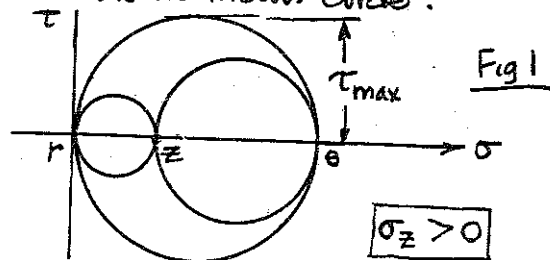
SUMMARY : $\tau_{r\theta} = \tau_{\theta z} = \tau_{zr} = 0 \quad \therefore r, \theta, z \equiv \text{principal axes}$
 $\sigma_r \approx 0$; $\sigma_\theta = \frac{pr}{2t}$; $\sigma_z = \frac{F}{2\pi r t}$

4.11 From prob 4.10 :

$\sigma_r \approx 0$
 $\sigma_\theta = \frac{pr}{2t}$
 $\sigma_z = \frac{F}{2\pi r t}$

shear stresses
 $= 0$

3 dimensional Mohr's circle :



4.11 cont'd

$\tau_{\max} = \frac{1}{2} |\sigma_{\max} - \sigma_{\min}| = \text{radius of}$
 a) largest circle.

1) if $F > 0$ then $\sigma_z > 0$

and $\tau_{\max} = \frac{1}{2} \sigma_z$ or $\frac{1}{2} \sigma_\theta$

whichever is greater (Fig 1 above)

Also $\sigma_{\max} = \sigma_z$ or σ_θ whichever

is greater. Hence $\tau_{\max}$ cannot possibly be equal to $\sigma_{\max}$.

2) Therefore $F < 0$: $\tau_{\max} = \frac{1}{2} (\sigma_\theta - \sigma_z) = \frac{1}{2} \left(\frac{pr}{t} - \frac{F}{2\pi r t} \right)$

$$\sigma_{\max} = \sigma_\theta = \frac{pr}{t}$$

$$\therefore \frac{1}{2} \left(\frac{pr}{t} - \frac{F}{2\pi r t} \right) = \frac{pr}{t}$$

ie.

$$F = -2\pi r^2 p$$

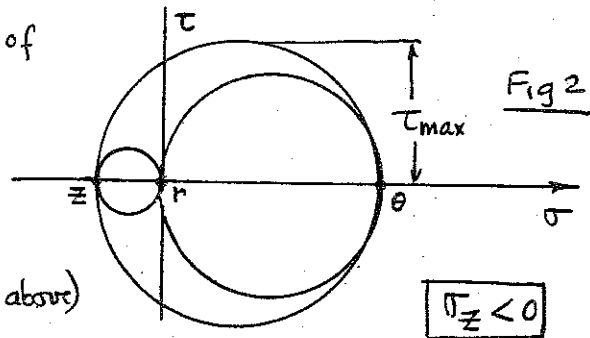
b) $\tau_{\max} = \frac{1}{2} \sigma_{\max}$.

1) $F > 0$, $\sigma_z > 0$ then $\tau_{\max}$ is always $\frac{1}{2} \sigma_{\max}$ (see above)

2) $F < 0$, $\sigma_z < 0$ $\frac{1}{2} \left(\frac{pr}{t} - \frac{F}{2\pi r t} \right) = \frac{1}{2} \left(\frac{pr}{t} \right)$ ie $F = 0$.

$\therefore$

$$F \geq 0$$



4.12 EQUILIBRIUM FOR FREE BODY

$$\sum F_z = 0 : F = p \pi r^2$$

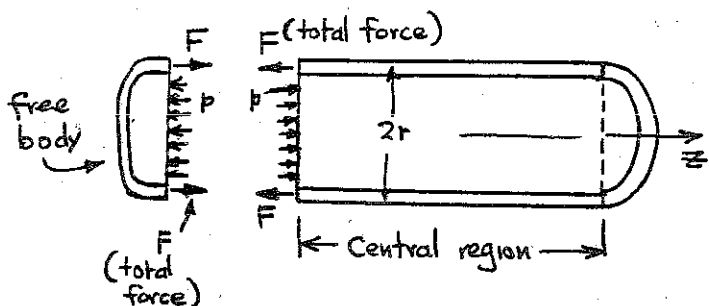
with a similar result for a free body made out of the other end of the cylinder.

Thus we may consider

the central region as a cylinder with internal pressure subjected to an axial force F . From prob 4.10 :-

$$\tau_{r\theta} = \tau_{\theta z} = \tau_{zr} = 0 \quad \therefore r, \theta, z \equiv \text{principal axes}$$

$$\sigma_r \approx 0 ; \sigma_\theta = \frac{pr}{2t} ; \sigma_z = \frac{F}{2\pi r t} = \frac{pr}{2t}$$



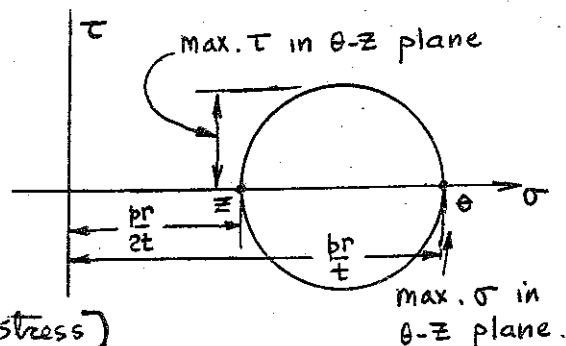
4.13 Prob 4.12 solution: principal stresses are $\begin{cases} \sigma_r = 0, \sigma_\theta = \frac{pr}{t} \\ \sigma_z = \frac{pr}{2t} \end{cases}$
 Mohr's circle in the θ - z plane:-
 (This is OK since $\tau_{r\theta} = \tau_{zr} = 0$)

$$(\tau_{\max})_{\theta z} = \frac{1}{2}(\sigma_\theta - \sigma_z) = \frac{pr}{4t}$$

$$(\sigma_{\max})_{\theta z} = \sigma_\theta = \frac{pr}{t}$$

$$\therefore (\tau_{\max})_{\theta z} = \frac{1}{4}(\sigma_{\max})_{\theta z}$$

[NOTE: $(\tau_{\max})_{\theta z}$ is not the max. shear stress]



4.14 Symmetry arguments similar to those used in prob 4.10 can be used to show that r, ϕ, θ are principal axes of stress.

EQUILIBRIUM

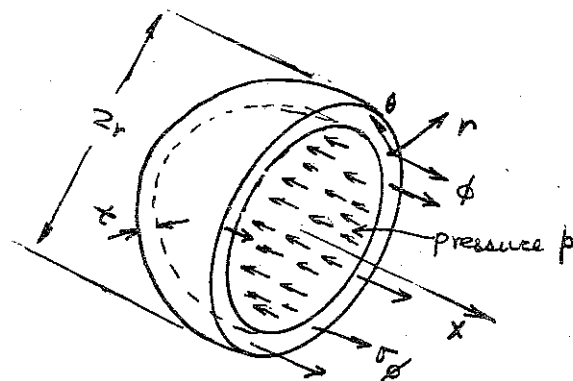
$$\sum F_x = 0: -p\pi r^2 + \sigma_\phi(2\pi r t) = 0$$

$$\therefore \sigma_\phi = \frac{pr}{2t}$$

Similarly $\sigma_\theta = \frac{pr}{2t}$

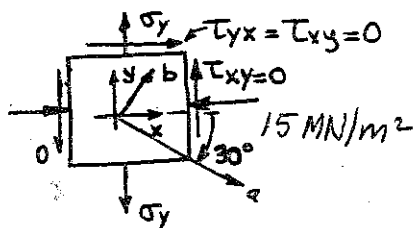
Inside surface $\sigma_r = -p$
 Outside surface $\sigma_r = 0$.

$\therefore$ reasonable to assume $-p \leq \sigma_r \leq 0$ everywhere. If $r/t \gg 1$ then $\sigma_r \ll \sigma_\theta, \sigma_\phi$ and may be neglected.



Free body: hemisphere + fluid in it.

4.15



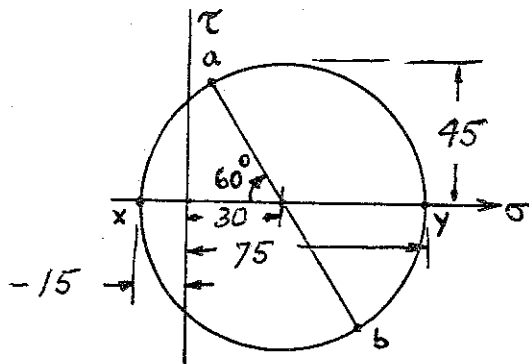
$$\sigma_{\max} = 75 \text{ MN/m}^2$$

The max. stress must be one of the principal stresses. Since there is no shear stress associated with the x - y axes, σ_x and σ_y must be principal stresses.

$$\sigma_x = -15 \text{ MN/m}^2, \therefore \sigma_y = 75 \text{ MN/m}^2$$

4.15 cont'd

MOHR'S CIRCLE : xy plane.

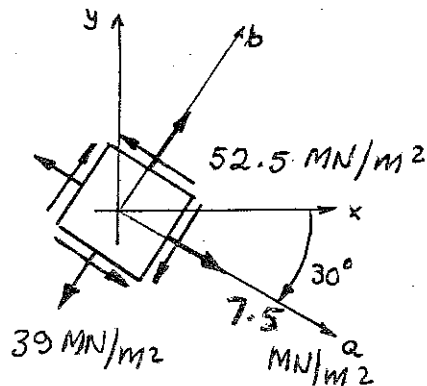


$$\sigma_a = 30 - 45 \cos 60^\circ$$

$$= 7.5 \text{ MN/m}^2$$

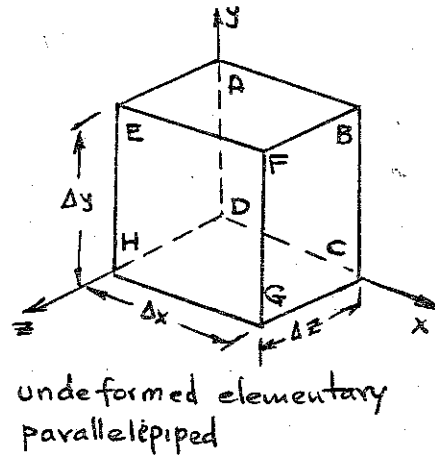
$$\sigma_b = 30 + 45 \cos 60^\circ = 52.5 \text{ MN/m}^2$$

$$\tau_{ab} = \text{NEGATIVE} = 38.97 \text{ MN/m}^2$$

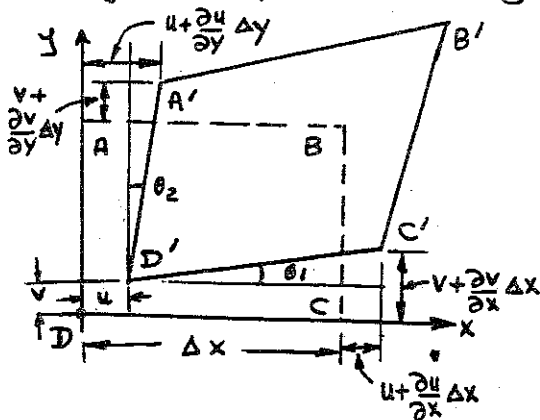


4.16

We will project three faces of the deformed parallelepiped on to the coordinate planes in order to obtain expressions for the strains. In this procedure we will take cosines of small angles as equal to unity and sines equal to the angle (also tangents = angle).



Projection of A'B'C'D' on x-y plane:



$$\epsilon_x = \frac{C'D' - CD}{CD} \approx \frac{(\Delta x + u + \frac{\partial u}{\partial x} \Delta x - u) - \Delta x}{\Delta x} = \frac{\partial u}{\partial x}$$

$$\epsilon_y = \frac{\partial v}{\partial y} \text{ in a similar fashion.}$$

$$\gamma_{xy} = \theta_1 + \theta_2 \approx \frac{(v + \frac{\partial v}{\partial x} \Delta x) - v}{\Delta x} + \frac{(u + \frac{\partial u}{\partial y} \Delta y) - u}{\Delta y}$$

$$= \left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \right)$$

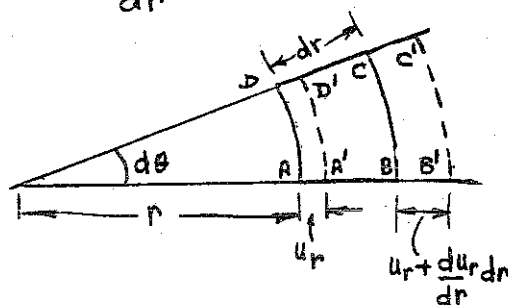
[Note: due to "w", the lines A'B' etc make a small angle with the x-y plane. The cosine is 1 and ∴ negligible effect on projection]

Similarly by projecting ADHE on the $y-z$ plane, and CDHG on the $z-x$ plane one obtains the corresponding results : $\epsilon_z = \frac{\partial w}{\partial z}$; $\gamma_{yz} = \frac{\partial w}{\partial y} + \frac{\partial v}{\partial z}$; $\gamma_{zx} = \frac{\partial u}{\partial z} + \frac{\partial w}{\partial x}$

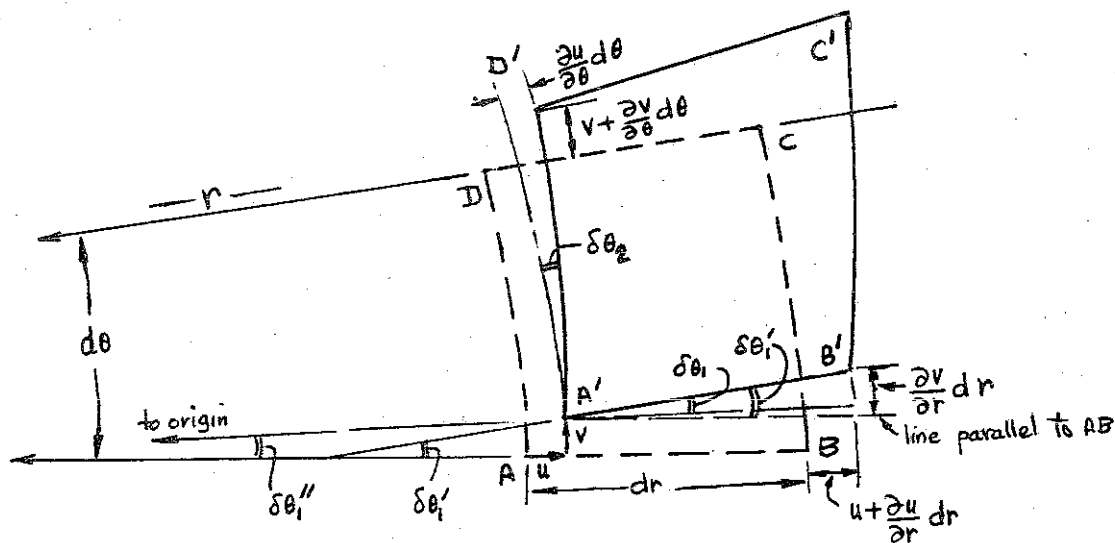
4.17 $\epsilon_r = \frac{A'B' - AB}{AB} = \frac{(dr + u_r + \frac{du_r}{dr} dr - u_r) - dr}{dr} = \frac{du_r}{dr}$

$\epsilon_\theta = \frac{A'D' - AD}{AD} = \frac{(r + u_r)d\theta - rd\theta}{rd\theta} = \frac{u_r}{r}$

$\gamma_{r\theta}$ = angle change between AB and AD = 0.



4.18



$\epsilon_r = \frac{A'B' - AB}{AB} = \frac{[(dr + u + \frac{\partial u}{\partial r} dr) - u] - [dr]}{dr} = \frac{\partial u}{\partial r}$

$\epsilon_\theta = \frac{A'D' - AD}{AD} = \frac{[(r + u)d\theta + v + \frac{\partial v}{\partial \theta} d\theta - r] - [rd\theta]}{rd\theta} = \frac{u}{r} + \frac{1}{r} \frac{\partial v}{\partial \theta}$

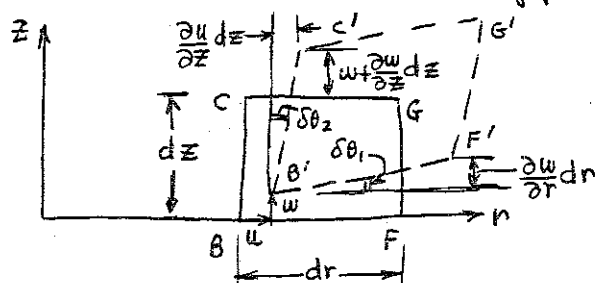
$\gamma_{r\theta} = \delta\theta_1 + \delta\theta_2 = (\delta\theta_1' - \delta\theta_1'') + \delta\theta_2 = \left(\frac{\partial v}{\partial r} dr - \frac{v}{r}\right) + \frac{\partial u}{\partial \theta} \frac{d\theta}{rd\theta} = \frac{\partial v}{\partial r} - \frac{v}{r} + \frac{1}{r} \frac{\partial u}{\partial \theta}$

4.19 By projecting $B'F'E'A'$ on the $r-\theta$ plane we obtain, as in prob 4.18,

$$\epsilon_r = \frac{\partial u}{\partial r}; \epsilon_\theta = \frac{1}{r} \frac{\partial v}{\partial \theta} - \frac{u}{r}$$

$$\gamma_{r\theta} = \frac{\partial v}{\partial r} + \frac{1}{r} \frac{\partial u}{\partial \theta} - \frac{v}{r}$$

$B'F'E'A'$ projected on a plane parallel to $BFGC$. The angle is small and the cosines approximately equal unity.



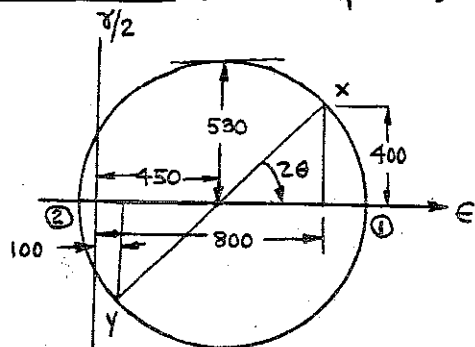
$$\begin{aligned} \epsilon_z &= \frac{B'C' - BC}{BC} \\ &= \frac{[dz + w + \frac{\partial w}{\partial z} dz - w] - dz}{dz} \\ &= \frac{\partial w}{\partial z} \end{aligned}$$

$$\gamma_{zr} = \delta\theta_1 + \delta\theta_2 = \frac{\partial w}{\partial r} dr + \frac{\partial u}{\partial z} dz = \frac{\partial w}{\partial r} + \frac{\partial u}{\partial z}$$

Similarly, by projecting $A'B'C'D'$ on the plane passing through points A, B, C and D, we obtain: $\gamma_{\theta z} = \frac{\partial v}{\partial z} + \frac{1}{r} \frac{\partial w}{\partial \theta}$

4.20 $\epsilon_x = 800 \times 10^{-6}$; $\epsilon_y = 100 \times 10^{-6}$; $\gamma_{xy} = -800 \times 10^{-6}$

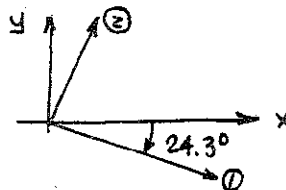
Mohr's Circle ($\epsilon \times 10^6$ is plotted)



$$\theta = \frac{1}{2} \tan^{-1} \left(\frac{400}{350} \right) = 24.3^\circ$$

$$\epsilon_1 = [450 + 530] 10^{-6} = 980 \times 10^{-6}$$

$$\epsilon_2 = [450 - 530] 10^{-6} = -80 \times 10^{-6}$$



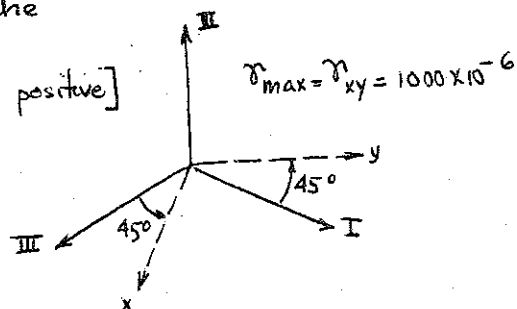
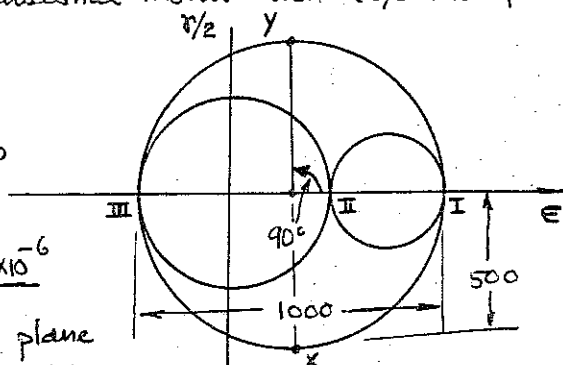
4.21 $\epsilon_I = 700 \times 10^{-6}$ Three dimensional Mohr's circle ($\epsilon, \gamma \times 10^6$ plotted)
 $\epsilon_{II} = 300 \times 10^{-6}$
 $\epsilon_{III} = -300 \times 10^{-6}$

Max. shear strain corresponds to point y

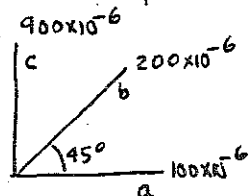
$$\frac{\gamma_{max}}{2} = 500 \times 10^{-6} \therefore \gamma_{max} = 1000 \times 10^{-6}$$

The y axis is in the I-III plane rotated clockwise by 45° from the I axis

[Note: y is anticlockwise axis $\therefore \gamma_{xy}$ is positive]



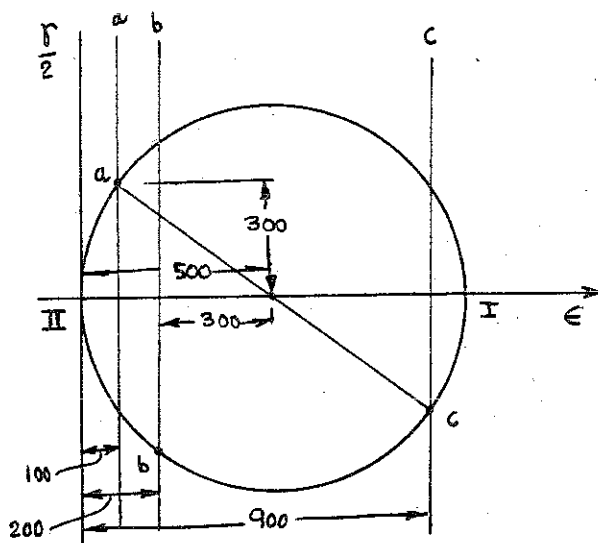
4.22 The procedure for drawing the Mohr's circle is outlined in Example 4.3 (strains $\times 10^6$ are plotted)



Thus

$$\epsilon^I = 1000 \times 10^{-6}$$

$$\epsilon^{II} = 0$$



(4-22) (b) $c = 60 \times 10^{-6}$ (Rosette readings)
 $b = 400 \times 10^{-6}$
 $a = 1200 \times 10^{-6}$

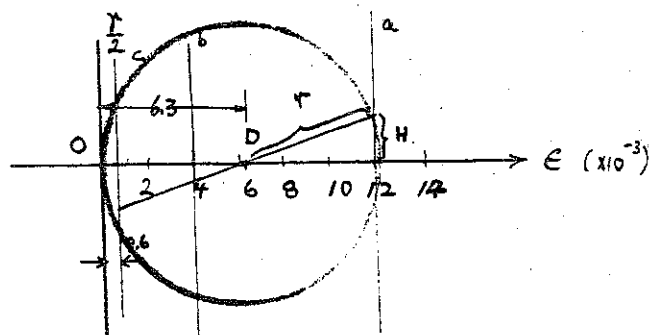
$$OD = \frac{1200 + 60}{2} = 630$$

$$H = 630 - 400 = 230$$

$$\therefore r = [(230)^2 + (570)^2]^{1/2} = 620$$

$$\epsilon_I = (630 + 620) = 1250 \times 10^{-6}$$

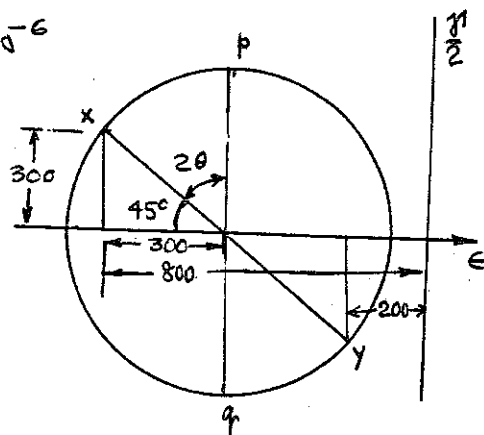
$$\epsilon_{II} = (630 - 620) = 10 \times 10^{-6}$$



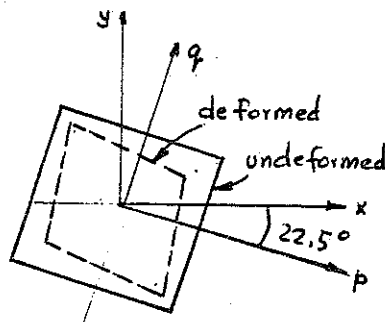
4.23 $\epsilon_x = -800 \times 10^{-6}$
 $\epsilon_y = -200 \times 10^{-6}$
 $\tau_{xy} = -600 \times 10^{-6}$

Mohr's circle :

$2\theta = 45^\circ$
 $\theta = 22.5^\circ$



p, q are max. shear axes.
 $\tau_{pq} = \text{negative}$

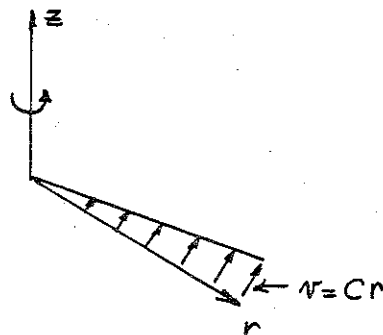


4.24 $u = w = 0 ; v = Cr$

$\epsilon_r = \frac{\partial u}{\partial r} = 0 ; \epsilon_\theta = \frac{1}{r} \frac{\partial v}{\partial \theta} + \frac{u}{r} = 0 ; \epsilon_z = \frac{\partial w}{\partial z} = 0 ;$

$\tau_{r\theta} = \frac{\partial v}{\partial r} + \frac{1}{r} \frac{\partial u}{\partial \theta} - \frac{v}{r} = C + 0 - C = 0 ; \tau_{\theta z} = \frac{1}{r} \frac{\partial w}{\partial \theta} + \frac{\partial v}{\partial z} = 0 ; \tau_{zr} = \frac{\partial u}{\partial z} + \frac{\partial w}{\partial r} = 0 .$

The motion is a rotation about the z axis.



4.25 GEOMETRY : $2\pi r \cos \alpha = w$

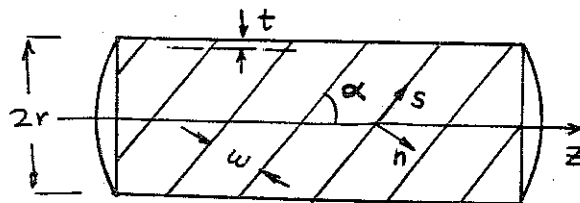
From prob 4.12 :

$\sigma_r \sim 0$

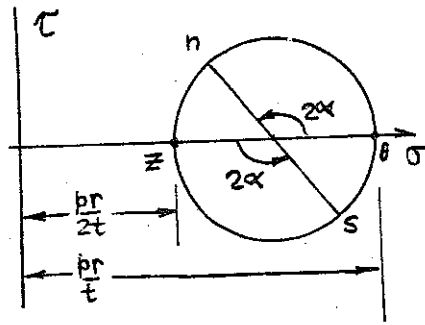
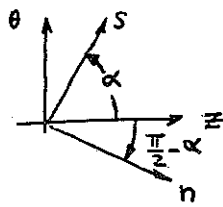
$\sigma_\theta = \frac{pr}{t}$

$\sigma_z = \frac{pr}{2t}$

shear stresses = 0



Use Mohr's circle to transform to $n-s$ axes (see figure)
 $(\theta = z \text{ plane})$



$$\sigma_n = \frac{1}{2} \left(\frac{pr}{2t} + \frac{pr}{t} \right) + \frac{1}{2} \left(\frac{pr}{t} - \frac{pr}{2t} \right) \cos 2\alpha$$

$$= \frac{3}{4} \left(\frac{pr}{t} \right) + \frac{1}{4} \left(\frac{pr}{t} \right) \cos 2\alpha$$

σ_n is the tensile stress across the fused joint.

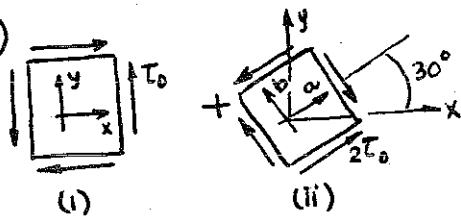
$\sigma_{max} = \sigma_{\theta} = \frac{pr}{t}$ if $\sigma_n = 0.8 \sigma_{max}$:

$$\frac{3}{4} \left(\frac{pr}{t} \right) + \frac{1}{4} \left(\frac{pr}{t} \right) \cos 2\alpha = 0.8 \left(\frac{pr}{t} \right) \quad \text{i.e.} \quad \cos 2\alpha = \frac{0.05}{0.25} = 0.20$$

$\theta = 39.2^\circ$; $\cos \alpha = 0.775$; $w = 2\pi r \cos \alpha$. $\therefore \left(\frac{w}{r} \right) = 4.86$

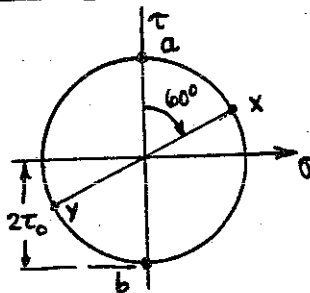
Thus $\left(\frac{w}{r} \right) \leq 4.86$

4.26 a)

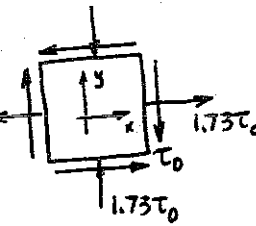


Using Mohr's circle, transform (ii) to x-y axes, then add to (i). Finally, find the principal axes.

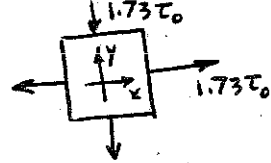
Mohr's circle for (ii) :



$$\left. \begin{aligned} \sigma_x &= 2\tau_0 \sin 60^\circ \\ &= 1.73\tau_0 \\ \sigma_y &= -1.73\tau_0 \\ \tau_{xy} &= -\tau_0 \end{aligned} \right\}$$

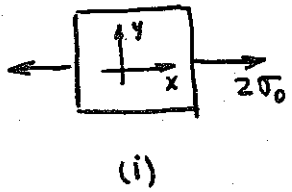


Add this stress state to (i). This results in

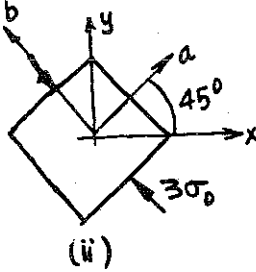


i.e. x, y are principal stress directions.

b)

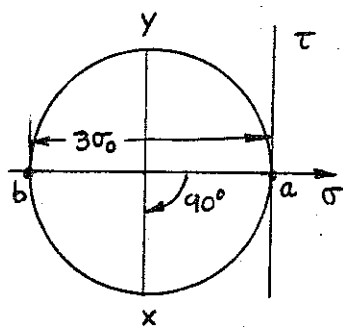


+

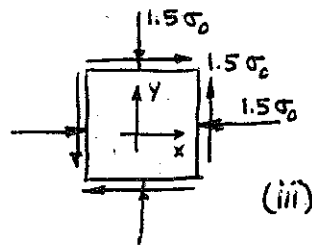


Proceed as in (a)

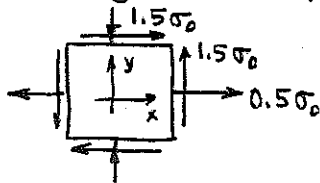
4.26 cont'd



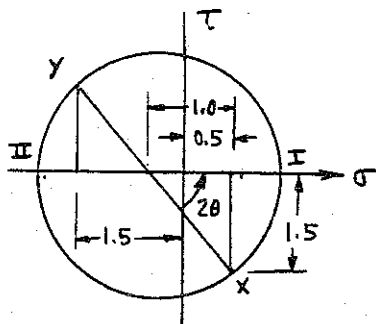
$$\left. \begin{aligned} \sigma_x &= -\frac{3}{2}\sigma_0 \\ \sigma_y &= -\frac{3}{2}\sigma_0 \\ \tau_{xy} &= \frac{3}{2}\sigma_0 \end{aligned} \right\}$$



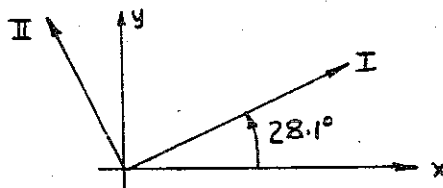
Adding (iii) to (i), we get



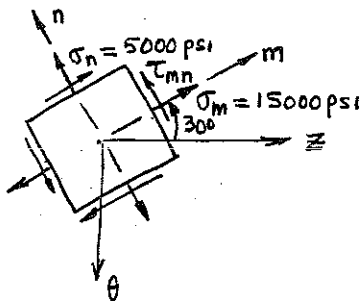
Mohr's Circle: (σ_0) is plotted



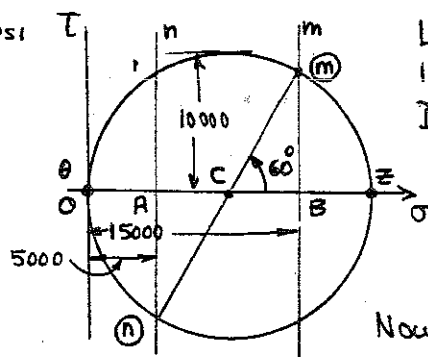
$$\theta = \frac{1}{2} \tan^{-1} \left(\frac{1.5}{-1.0} \right) = 28.1^\circ$$



4.27 a)



Mohr's circle construction:



Locate Band A at $\sigma = 15000$ and 5000 respectively. Draw vertical lines through them. Locate the center C midway between A and B.

Now, using prob 4.10,

$\sigma_z = \frac{F}{2\pi r t}$, $\tau_\theta = \frac{p r}{t}$ and θ, z are principal axes of stress. Points corresponding to θ, z therefore lie on the σ axis ($\tau=0$). Draw lines at $\pm 60^\circ$ to the σ axis from C, to locate (m) and (n). The sign of the angle should be chosen so that z, m, θ, n are in anticlockwise order,

4.27 cont'd

and Ξ is 60° clockwise from m . This results in the figure shown.

$$\sigma_\theta = 0; \therefore \underline{p=0} \quad \sigma_\Xi = 20,000 \text{ psi} = \frac{F}{2\pi r t}$$

$$\therefore F = (20,000)(2\pi)(10)(0.1) = \underline{126,000 \text{ lb}}$$

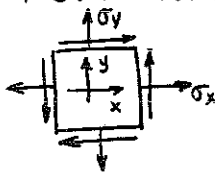
$$\text{Also, } \tau_{mn} \text{ is negative and } = -10,000 \sin 60^\circ = \underline{-8670 \text{ psi}}$$

b) Mohr's circle shrinks to a point.

$$\sigma_\theta = \sigma_\Xi = 15000 \text{ psi} \text{ and } \underline{\tau_{mn} = 0}$$

$$p = \left(\frac{\pm}{r}\right)\sigma_\theta = \left(\frac{0.1}{10}\right)15000 = \underline{150 \text{ psi}}; F = (2\pi)(10)(0.1)15000 = \underline{94000 \text{ lb}}$$

4.28 Mohr's circle construction:



20 MN/m^2

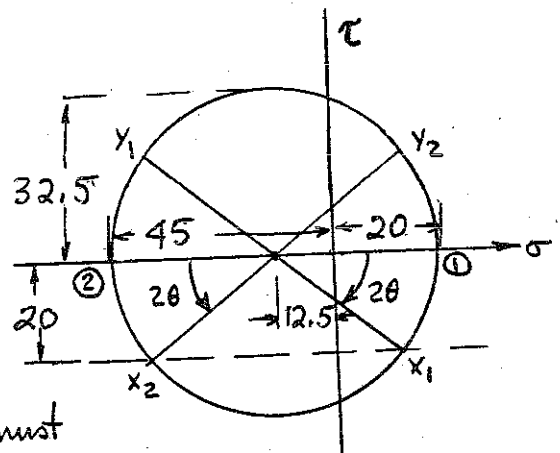
Locate ①, ② and the center.

Since τ_{xy} is positive and x is the clockwise axis

the point corresponding to x must

lie on the lower half of the circle. Draw a line at

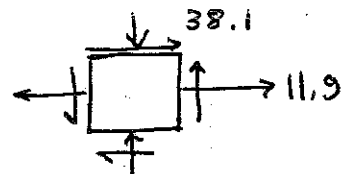
$\tau = 20 \text{ MN/m}^2$. This intersects the circle at x_1 and x_2 and there is no way of deciding which is the right one. There are therefore two possible stress states.



$$(32.5) \sin 2\theta = 20 \quad \theta = 19^\circ \quad \sin 2\theta = 0.615 \quad \cos 2\theta = 0.788$$

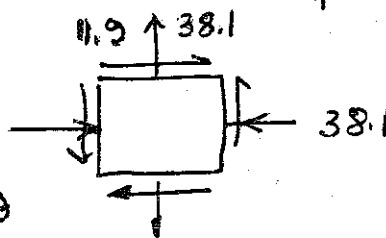
$$(x_1, y_1): \sigma_x = -12.5 + 32.5 \cos 2\theta = 11.9 \text{ MN/m}^2$$

$$\sigma_y = -38.1 \text{ MN/m}^2$$

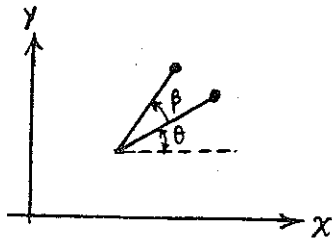


$$(x_2, y_2): \sigma_x = -38.1 \text{ MN/m}^2$$

$$\sigma_y = 11.9 \text{ MN/m}^2$$



(4-29)



displacement of rigid body rotation is :

$$u(x, y) = (\cos \beta - 1)x - \sin \beta y$$

$$v(x, y) = \sin \beta x + (\cos \beta - 1)y$$

From eq. (4.32) $w_z = \frac{1}{2} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) = \frac{1}{2} (\sin \beta + \sin \beta) = \sin \beta$

For small β , $\sin \beta \cong \beta$, $\therefore w_z \cong \beta$

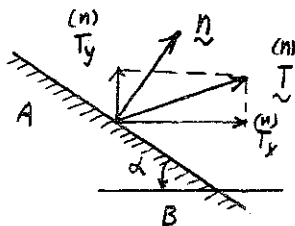
From eq. (4.33) $\epsilon_x = \frac{\partial u}{\partial x}$; $\epsilon_y = \frac{\partial v}{\partial y}$; $\gamma_{xy} = \frac{\partial v}{\partial x} + \frac{\partial u}{\partial y}$

$$\therefore \epsilon_x = \cos \beta - 1 \quad \left. \begin{array}{l} \epsilon_y = \cos \beta - 1 \\ \gamma_{xy} = \sin \beta - \sin \beta = 0 \end{array} \right\} \text{ for } \beta \ll 1, \quad \begin{array}{l} \epsilon_x = O(\beta^2) \\ \epsilon_y = O(\beta^2) \end{array}$$

$$\epsilon_y = \cos \beta - 1$$

$$\gamma_{xy} = \sin \beta - \sin \beta = 0$$

(4-30)



From equilibrium in general

$$\frac{(n)}{T_i} = \tau_{ij} n_j$$

For our 2-D case, stress free case

$$\frac{(n)}{T_x} = \tau_{xx} n_x + \tau_{xy} n_y = 0 \quad (1)$$

$$\frac{(n)}{T_y} = \tau_{xy} n_x + \tau_{yy} n_y = 0 \quad (2)$$

$$\underline{n} = \sin \alpha \underline{i} + \cos \alpha \underline{j} \quad \text{or } n_x = \sin \alpha$$

$$n_y = \cos \alpha$$

$\therefore$ (1) & (2) becomes

$$\tau_{xx} \sin \alpha + \tau_{xy} \cos \alpha = 0$$

$$\tau_{xy} \sin \alpha + \tau_{yy} \cos \alpha = 0$$

(4-31) From eq. (4-25)

$$\sigma_x' = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2} \cos 2\theta + \tau_{xy} \sin 2\theta$$

For σ_x' to be max. or min.

$$\frac{\partial \sigma_x'}{\partial \theta} = 0 = -2 \frac{\sigma_x - \sigma_y}{2} \sin 2\theta + 2 \tau_{xy} \cos 2\theta \rightarrow \frac{\sigma_x - \sigma_y}{2} \sin 2\theta = \tau_{xy} \cos 2\theta \dots (1)$$

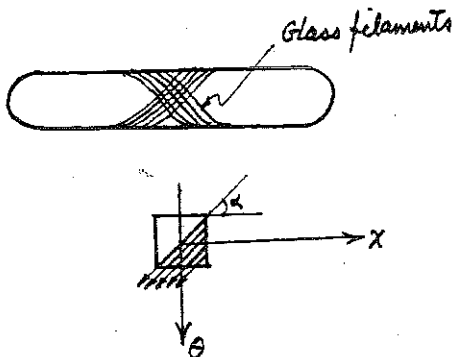
From second eq. of (4-25)

$$\tau_{x'y'} = -\frac{\sigma_x - \sigma_y}{2} \sin 2\theta + \tau_{xy} \cos 2\theta$$

$$\text{From (1), } \cos 2\theta = \frac{\sigma_x - \sigma_y}{2 \tau_{xy}} \quad \therefore \tau_{xy} = -\frac{\sigma_x - \sigma_y}{2} \sin \theta + \frac{\sigma_x - \sigma_y}{2} \sin \theta = 0$$

$\therefore$ shear stress τ_{xy} vanishes on the plane defined by θ & this plane is the plane of principal axes.

(4-32)



Assume the stress in the θ direction is the force in the filament divided by the area.
Let F equal to the force in a filament

$$\frac{n F \sin \alpha}{1 \cdot t} = \sigma_\theta = \frac{p r}{t}$$

$$\frac{n F \cos \alpha}{t \cdot 1 \cdot \tan \alpha} = \sigma_x = \frac{p r}{2 t}$$

$$\therefore \tan^2 \alpha = 2 \rightarrow \tan \alpha = 1.414$$

$$\alpha \approx 55^\circ$$

(4.33) Equilibrium equation for slender beam, from eq. (3-11) & (3-12)

$$\frac{dV}{dx} + q = 0 ; \quad \frac{dM_b}{dx} + V = 0$$

Integrating eq. (4-13) across thickness of beam

$$\int_{-h/2}^{h/2} \frac{\partial \tau_{xy}}{\partial x} dy + \int_{-h/2}^{h/2} \frac{\partial \sigma_y}{\partial y} dy = 0 \rightarrow \frac{\partial}{\partial x} \int_{-h/2}^{h/2} \tau_{xy} dy + \left. \sigma_y \right|_{y=h/2} - \left. \sigma_y \right|_{y=-h/2} = 0$$

$$V = \int_{-h/2}^{h/2} \tau_{xy} dy \quad q = \left. \sigma_y \right|_{y=h/2} - \left. \sigma_y \right|_{y=-h/2}$$

$$\therefore \frac{\partial V}{\partial x} + q = 0$$

Multiplying the 1st of (4-13) and integrate across thickness

$$\int_{-h/2}^{h/2} y \frac{\partial \sigma_x}{\partial x} dy + \int_{-h/2}^{h/2} y \frac{\partial \tau_{xy}}{\partial y} dy = 0 \rightarrow \frac{\partial}{\partial x} \int_{-h/2}^{h/2} y \sigma_x dy + \left. y \tau_{xy} \right|_{-h/2}^{h/2} - \int_{-h/2}^{h/2} \tau_{xy} dy = 0$$

$$\text{But } \left. \tau_{xy} \right|_{-h/2} = - \left. \tau_{xy} \right|_{h/2} \therefore \left. y \tau_{xy} \right|_{-h/2}^{h/2} = 0$$

$$\therefore - \frac{\partial}{\partial x} \int_{-h/2}^{h/2} y \sigma_x dy + \int_{-h/2}^{h/2} \tau_{xy} dy = 0 \rightarrow \frac{dM}{dx} + V = 0$$

(4.34) Consider sea-urchin egg to be thin wall sphere.

$$\text{Volume} = \frac{4}{3} \pi r^3 = 35 \times 10^{-9} \text{ cm}^3 \quad \therefore r^3 = \frac{3 \times 35 \times 10^{-9}}{4 \pi} = 83 \times 10^{-9} \text{ cm}^3$$

$$r = 4.37 \times 10^{-3} \text{ cm.}$$

$$\text{Internal pressure } P_i = 150 \text{ mmHg} \\ = 2 \times 10^5 \text{ dynes/cm}^2$$

$$\text{thickness of membrane } t = 10^{-4} \text{ cm.}$$

$$\therefore \sigma = \frac{rP}{2t} = \frac{4.37 \times 10^{-3} \times 2 \times 10^5}{2 \times 10^{-4}} = 4.37 \times 10^6 \text{ dynes/cm}^2$$

$$1 \text{ dyne} = 10^{-5} \text{ N}$$